

Project B14: Student Task & Portfolio Repository

Project Statement

Build a centralized digital repository where all student work — projects, assignments, submissions, practical tasks, and evaluations — is stored, organized, and made accessible throughout the student's academic lifecycle. The system must support submission of various file types, version tracking per submission, teacher evaluation and grading within the platform, and AI-powered portfolio generation that compiles a student's best work into a shareable, professional portfolio document or microsite.

Expected Output

Frontend:

- Student submission portal: upload work per assignment/task with description and tags
- Submission history view: all past submissions with status (submitted, graded, returned)
- Teacher evaluation interface: view submissions, add rubric-based scores and written feedback
- Portfolio generation wizard: student selects best works, AI organizes and generates a portfolio
- Portfolio preview: shareable link to a clean, formatted portfolio page
- Admin/teacher view: all submissions per batch filtered by subject, date, or task type

Backend:

- File storage API: handles multiple file types (PDF, ZIP, images, videos, code files)
- Submission management APIs: create, update, retrieve submissions per student per task
- Evaluation API: rubric definition, scoring, and feedback storage
- Version tracking: stores all versions of a submission
- Portfolio generation AI engine: selects, organizes, and formats student work into a structured output
- Portfolio hosting: generates a static or dynamic shareable portfolio page per student

Deployment:

- Scalable file storage for thousands of students' submissions
- Portfolio pages should be publicly accessible via unique URL when shared
- Fine-grained access control: students own their work, teachers can grade, admins can audit

Note to interns: These projects are intentionally scoped at production-system complexity. You are expected to make all architectural, technical, and design decisions independently — including how to handle scale, latency, data privacy, model selection, and infrastructure. Ability to research, justify, and defend your choices is part of the evaluation.